

Bass Real Analysis Solutions

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Table of contents

1	Home	5
2	Bass Real Analysis Solutions	6
2.1	About this project	6
2.2	Author	6
3	About	7
4	Jae-Yun Lee	8
4.1	Education	8
4.2	Research	8
4.3	Music	8
I	Chapter 2	9
5	Exercise 2.1	10
5.1	Solution	10
6	Exercise 2.2	11
6.1	Solution	11
7	Exercise 2.3	12
7.1	Solution	12
8	Exercise 2.4	14
8.1	Solution	14
9	Exercise 2.5	15
9.1	Solution	15
10	Exercise 2.6	16
10.1	Solution	16
11	Exercise 2.7	17
11.1	Solution	17

12 Exercise 2.8	19
12.1 Solution	19
13 Exercise 2.9	21
13.1 Solution	21
 II Chapter 3	 24
14 Exercise 3.1	25
14.1 Solution	25
15 Exercise 3.2	26
15.1 Solution	26
16 Exercise 3.3	27
16.1 Solution	27
17 Exercise 3.4	28
17.1 Solution	28
18 Exercise 3.5	29
18.1 Solution	29
19 Exercise 3.6	30
19.1 Solution	30
20 Exercise 3.7	31
20.1 Solution	31
21 Exercise 3.8	32
21.1 Solution	32
22 Exercise 3.9	35
22.1 Solution	35
23 Exercise 3.10	38
23.1 Solution	38
 III Chapter 4	 39
24 Exercise 4.1	40
24.1 Problem	40
24.2 Solution	40

25 Exercise 4.2	41
25.1 Problem	41
25.2 Solution	41
26 Exercise 4.3	42
26.1 Problem	42
26.2 Solution	42
References	43

1 Home

2 Bass Real Analysis Solutions

This website contains my solutions to selected exercises from Richard Bass's *Real Analysis for Graduate Students v5.0*.

2.1 About this project

The goal of this project is to provide clear and rigorous solutions to graduate-level real analysis exercises.

2.2 Author

Jae-Yun Lee

- B.S. in Mathematical Sciences, KAIST
- M.S. and Ph.D. Track in Economics, KAIST

For more information, see the [About](#) page.

3 About

4 Jae-Yun Lee

I am an independent researcher and musician from South Korea.

4.1 Education

- B.S. in Mathematical Sciences, KAIST
- M.S. and Ph.D. Track in Economics, Department of Management Engineering, KAIST

4.2 Research

Selected publication:

Jae-Yun Lee, Jiwoong Shin, and Jungju Yu

[“Communicating Attribute Importance under Competition”](#)

Marketing Science, 45(3), 534–555, 2026.

4.3 Music

Outside academia, I work as an independent musician under the name **Jayhat**.

My work combines hip-hop, R&B, and classical music, with a particular interest in orchestral composition and piano performance.

- Instagram: [\(jayhat3210?\)](#)

Part I

Chapter 2

5 Exercise 2.1

5.1 Solution

Consider $X = \{1, 2, 3\}$ and $\mathcal{M} = \{\emptyset, \{1\}, \{2\}, \{3\}, X\}$. It is straightforward to verify that $\mathcal{M}$ is a monotone class. However, $\mathcal{M}$ is not a σ -algebra, since

$$\{1\} \cup \{2\} = \{1, 2\} \notin \mathcal{M}.$$

6 Exercise 2.2

6.1 Solution

Consider $X = \{1, 2, 3, 4\}$, and

$$\mathcal{A}_1 = \{\emptyset, \{1\}, \{2, 3, 4\}, X\}, \quad \mathcal{A}_2 = \{\emptyset, \{1, 2, 3\}, \{4\}, X\}.$$

It is straightforward to verify that $\mathcal{A}_1$ and $\mathcal{A}_2$ are σ -algebras. However, $\mathcal{A}_1 \cup \mathcal{A}_2$ is not a σ -algebra, since

$$\{1\} \cup \{4\} = \{1, 4\} \notin \mathcal{A}_1 \cup \mathcal{A}_2.$$

7 Exercise 2.3

7.1 Solution

Consider $X = \mathbb{N}$ and $\mathcal{A}_n = \sigma(\{\{1\}, \{2\}, \dots, \{n\}\})$.

Note that $\mathcal{A}_n$ is a σ -algebra by construction. Moreover, since

$$\{\{1\}, \{2\}, \dots, \{n\}\} \subset \{\{1\}, \{2\}, \dots, \{n+1\}\},$$

we have $\mathcal{A}_n \subset \mathcal{A}_{n+1}$.

First, we establish a useful property of $\mathcal{A}_n$.

For each $n \in \mathbb{N}$, define

$$T_n = \{n+1, n+2, \dots\} \quad \text{and} \quad \mathcal{B}_n = \{E \subset \mathbb{N} : T_n \subset E \text{ or } E \cap T_n = \emptyset\}.$$

We claim that $\mathcal{B}_n$ is a σ -algebra.

(1) First, $\emptyset \in \mathcal{B}_n$, since $\emptyset \cap T_n = \emptyset$.

(2) Second, suppose that $E \in \mathcal{B}_n$.

If $T_n \subset E$, then $E^c \cap T_n = \emptyset$. And if $E \cap T_n = \emptyset$, then $T_n \subset E^c$.

Thus, we have $E^c \in \mathcal{B}_n$.

(3) Third, suppose that $E_k \in \mathcal{B}_n$ for $k = 1, 2, \dots$.

If there exists k such that $T_n \subset E_k$, then $T_n \subset \bigcup_k E_k$. Thus, we have $\bigcup_k E_k \in \mathcal{B}_n$.

Otherwise, $E_k \cap T_n = \emptyset$ for all k . Hence,

$$\left(\bigcup_k E_k \right) \cap T_n = \bigcup_k (E_k \cap T_n) = \emptyset.$$

Therefore, $\mathcal{B}_n$ is a σ -algebra.

Next, since $\{1\}, \dots, \{n\} \in \mathcal{B}_n$, we conclude that

$$\mathcal{A}_n = \sigma\left(\{\{1\}, \{2\}, \dots, \{n\}\}\right) \subset \mathcal{B}_n.$$

Consequently, for every $E \in \mathcal{A}_n$, we have $E \cap T_n = \emptyset$ or $T_n \subset E$.

Using this property, we will show that $\bigcup_k \mathcal{A}_k$ is not a σ -algebra.

Consider the set

$$\{2, 4, 6, \dots\} = \bigcup_n \{2n\}.$$

For any $n \in \mathbb{N}$, we have

$$\{2, 4, 6, \dots\} \cap T_n \neq \emptyset \quad \text{and} \quad T_n \not\subset \{2, 4, 6, \dots\}.$$

Hence, $\{2, 4, 6, \dots\} \notin \mathcal{A}_n$ for any $n \in \mathbb{N}$, which implies that $\{2, 4, 6, \dots\} \notin \bigcup_k \mathcal{A}_k$.

On the other hand, $\{2n\} \in \mathcal{A}_{2n} \subset \bigcup_k \mathcal{A}_k$ for each $n \in \mathbb{N}$.

Therefore, $\bigcup_k \mathcal{A}_k$ is not closed under countable unions, and therefore is not a σ -algebra.

8 Exercise 2.4

8.1 Solution

(I think it is $A_j \in \mathcal{M}_j$ in exercise, not $A_j \in \mathcal{M}$)

Consider $A_n = \{2, 4, \dots, 2n\}$ and $\mathcal{M}_n = \{\emptyset, A_n, \mathbb{N}\}$. It is straightforward to verify that

- (1) $\mathcal{M}_n$ is a monotone class, and
- (2) $A_n \uparrow A = \{2, 4, 6, \dots\}$.

However, $A \neq A_n$ for every n , so, $A \notin \mathcal{M}$.

9 Exercise 2.5

9.1 Solution

We show that $\mathcal{B}$ is a σ -algebra.

(1) $\emptyset \in \mathcal{B}$

Since $\emptyset \in \mathcal{A}$, we have $f^{-1}(\emptyset) \in \mathcal{B}$. And we can say that $f^{-1}(\emptyset) = \emptyset$.

To prove this, suppose not. Then there exists $x \in X$, such that $x \in f^{-1}(\emptyset)$. That is, $f(x) \in \emptyset$. But it is a contradiction.

(2) $X \in \mathcal{B}$

Since $Y \in \mathcal{A}$, we have $f^{-1}(Y) \in \mathcal{B}$. And we can say that $f^{-1}(Y) = X$.

To prove this, suppose not. Then there exists $x \in X$, such that $x \notin f^{-1}(Y)$. That is, $f(x) \notin Y$. But it is a contradiction.

(3) If $B \in \mathcal{B}$, then $B^c \in \mathcal{B}$

Consider $B \in \mathcal{B}$. Then there exists $A \in \mathcal{A}$, such that $f^{-1}(A) = B$. Since

$$f^{-1}(A^c) = X \setminus f^{-1}(A) = B^c,$$

and $A^c \in \mathcal{A}$, we can conclude that $B^c \in \mathcal{B}$.

(4) If $B_1, B_2, \dots \in \mathcal{B}$, then $\bigcup_i B_i \in \mathcal{B}$

Consider $B_1, B_2, \dots \in \mathcal{B}$ and corresponding $A_1, A_2, \dots \in \mathcal{A}$. Since

$$\bigcup_i B_i = \bigcup_i f^{-1}(A_i) = f^{-1}\left(\bigcup_i A_i\right),$$

and $\bigcup_i A_i \in \mathcal{A}$, we can conclude that $\bigcup_i B_i \in \mathcal{B}$.

10 Exercise 2.6

10.1 Solution

Suppose, for contradiction, that $\mathcal{A}$ is finite.

Since $X \neq \emptyset$ and $X \in \mathcal{A}$, the assumed property implies that X can be written as a disjoint union of two nonempty sets in $\mathcal{A}$.

Choose one of them and call it A_1 . Now apply the same property to A_1 . Then A_1 can be written as a disjoint union of two nonempty sets in $\mathcal{A}$.

Choose one of them and call it A_2 . Since the other piece is also nonempty, we have

$$A_2 \subsetneq A_1.$$

Repeating this argument, we obtain a strictly decreasing sequence

$$X \supsetneq A_1 \supsetneq A_2 \supsetneq A_3 \supsetneq \dots$$

of sets in $\mathcal{A}$.

This is impossible if $\mathcal{A}$ is finite. Hence $\mathcal{A}$ is infinite.

By Exercise 2.8, every infinite σ -algebra is uncountable. Therefore, $\mathcal{A}$ is uncountable.

11 Exercise 2.7

11.1 Solution

We show that $\mathcal{A}$ is a σ -algebra.

(1) $\emptyset \in \mathcal{A}$

Since $\chi_{\emptyset}(x) = 0$ for every $x \in X$, it is a constant function. Hence,

$$\chi_{\emptyset} \in \mathcal{F}.$$

Therefore, $\emptyset \in \mathcal{A}$.

(2) $X \in \mathcal{A}$

Since $\chi_X(x) = 1$ for every $x \in X$, it is a constant function. Hence,

$$\chi_X \in \mathcal{F}.$$

Therefore, $X \in \mathcal{A}$.

(3) If $A \in \mathcal{A}$, then $A^c \in \mathcal{A}$

Let $A \in \mathcal{A}$. Then $\chi_A \in \mathcal{F}$. Since

$$\chi_{A^c} = 1 - \chi_A,$$

and since $1 \in \mathcal{F}$, it follows that

$$\chi_{A^c} \in \mathcal{F}.$$

Therefore, $A^c \in \mathcal{A}$.

(4) If $A_1, A_2, \dots \in \mathcal{A}$, then $\bigcup_i A_i \in \mathcal{A}$

First, if $\chi_A, \chi_B \in \mathcal{F}$, then

$$\chi_{A \cup B} = \chi_A + \chi_B - \chi_A \chi_B.$$

Since $\chi_A, \chi_B \in \mathcal{F}$, we have

$$\chi_{A \cup B} \in \mathcal{F}.$$

Thus, for every $n \in \mathbb{N}$,

$$f_n := \chi_{\bigcup_{i=1}^n A_i} \in \mathcal{F}.$$

Moreover,

$$f_n(x) \rightarrow \chi_{\bigcup_i A_i}(x) \quad \text{for every } x \in X.$$

Since $\mathcal{F}$ is closed under pointwise limits, we have

$$\chi_{\bigcup_i A_i} \in \mathcal{F}.$$

Therefore,

$$\bigcup_i A_i \in \mathcal{A}.$$

Hence, $\mathcal{A}$ is a σ -algebra.

12 Exercise 2.8

12.1 Solution

Let X be a set and let $\mathcal{A}$ be a σ -algebra on X . Suppose that $\mathcal{A}$ is infinite. Then X must also be infinite, since if X were finite, then $\mathcal{A} \subset \mathcal{P}(X)$ would also be finite.

We will show that $\mathcal{A}$ has uncountably many elements. For this, we first construct a sequence of pairwise disjoint nonempty sets $B_1, B_2, \dots \in \mathcal{A}$. Then, we show that $\mathcal{P}(\mathbb{N})$ injects into $\mathcal{A}$.

Step 1:

Claim. For $E \in \mathcal{A}$, if $\{E \cap A : A \in \mathcal{A}\}$ is infinite, then there exists $C \in \mathcal{A}$ such that

$$\emptyset \neq E \cap C \subsetneq E.$$

Proof. Suppose, to the contrary, that no such C exists. Then for every $A \in \mathcal{A}$, we have

$$E \cap A = \emptyset \quad \text{or} \quad E \cap A = E.$$

Hence,

$$\{E \cap A : A \in \mathcal{A}\} = \{\emptyset, E\},$$

which contradicts the assumption that this collection is infinite. ■

Now let $A_1 = X$. Then

$$\{A_1 \cap A : A \in \mathcal{A}\} = \mathcal{A}$$

is infinite.

Suppose that $A_n \in \mathcal{A}$ has been chosen so that

$$\{A_n \cap A : A \in \mathcal{A}\}$$

is infinite. By the claim, there exists $C \in \mathcal{A}$ such that

$$\emptyset \neq A_n \cap C \subsetneq A_n.$$

Then both $A_n \cap C$ and $A_n \cap C^c$ are nonempty members of $\mathcal{A}$.

At least one of the two collections

$$\{(A_n \cap C) \cap A : A \in \mathcal{A}\} \quad \text{and} \quad \{(A_n \cap C^c) \cap A : A \in \mathcal{A}\}$$

must be infinite; otherwise $\{A_n \cap A : A \in \mathcal{A}\}$ would be finite.

Choose A_{n+1} to be one of $A_n \cap C$ and $A_n \cap C^c$ whose associated collection is infinite.

By induction, we obtain a strictly decreasing sequence

$$A_1 \supsetneq A_2 \supsetneq \dots$$

of nonempty sets in $\mathcal{A}$.

Define

$$B_n = A_n \setminus A_{n+1}.$$

Then each B_n is a nonempty member of $\mathcal{A}$, and the sets $B_1, B_2, \dots$ are pairwise disjoint.

Step 2:

For each $N \subset \mathbb{N}$, define

$$E_N = \bigcup_{n \in N} B_n.$$

Since $\mathcal{A}$ is a σ -algebra, we have $E_N \in \mathcal{A}$.

Moreover, since the sets B_n are pairwise disjoint and nonempty, if $N \neq M$, then

$$E_N \neq E_M.$$

Thus the map

$$N \mapsto E_N$$

is an injection from $\mathcal{P}(\mathbb{N})$ into $\mathcal{A}$.

Since $\mathcal{P}(\mathbb{N})$ is uncountable, $\mathcal{A}$ is uncountable.

13 Exercise 2.9

13.1 Solution

(1) Let $x \in \liminf_i A_i$. Then there exists $j \in \mathbb{N}$ such that

$$x \in \bigcap_{i=j}^{\infty} A_i.$$

Thus $x \in A_i$ for every $i \geq j$, and hence $x \in A_i$ for all but finitely many i .

Conversely, let $x \in A_i$ for all but finitely many i . Then there exists i^* such that $x \in A_i$ for every $i \geq i^*$. Hence,

$$x \in \bigcap_{i=i^*}^{\infty} A_i \subset \liminf_i A_i.$$

Therefore,

$$\liminf_i A_i = \{x : x \in A_i \text{ for all but finitely many } i\}.$$

Let $x \in \limsup_i A_i$. Then for every $j \in \mathbb{N}$, we have

$$x \in \bigcup_{i=j}^{\infty} A_i.$$

But if $x \in A_i$ for only finitely many i , then there exists i^* such that $x \notin A_i$ for every $i \geq i^*$. It follows that

$$x \notin \bigcup_{i=i^*}^{\infty} A_i,$$

which is a contradiction. Therefore, $x \in A_i$ for infinitely many i .

Conversely, let $x \in A_i$ for infinitely many i . Then for every j , we have

$$x \in \bigcup_{i=j}^{\infty} A_i.$$

Hence, we have

$$x \in \bigcap_{j=1}^{\infty} \bigcup_{i=j}^{\infty} A_i.$$

Therefore,

$$\limsup_i A_i = \{x : x \in A_i \text{ for infinitely many } i\}.$$

(2) Define

$$A_n = \left(-\frac{1}{n}, 1\right] \text{ if } n \text{ is odd, } A_n = \left(-1, \frac{1}{n}\right] \text{ if } n \text{ is even.}$$

For every n , we have

$$\bigcup_{i=n}^{\infty} A_i = (A_n \cup A_{n+1}) \cup (A_{n+2} \cup A_{n+3}) \cup \dots = (-1, 1] \cup (-1, 1] \cup \dots = (-1, 1].$$

Therefore,

$$\limsup_i A_i = \bigcap_{j=1}^{\infty} \bigcup_{i=j}^{\infty} A_i = (-1, 1].$$

If n is odd, we have

$$\begin{aligned} \bigcap_{i=n}^{\infty} A_i &= (A_n \cap A_{n+1}) \cap (A_{n+2} \cap A_{n+3}) \cap \dots \\ &= \left(-\frac{1}{n}, \frac{1}{n+1}\right] \cap \left(-\frac{1}{n+2}, \frac{1}{n+3}\right] \cap \dots = \{0\}. \end{aligned}$$

And if n is even, we have

$$\bigcap_{i=n}^{\infty} A_i = \left(-\frac{1}{n+1}, \frac{1}{n}\right] \cap \left(-\frac{1}{n+3}, \frac{1}{n+2}\right] \cap \dots = \{0\}.$$

Therefore,

$$\liminf_i A_i = \bigcup_{j=1}^{\infty} \bigcap_{i=j}^{\infty} A_i = \{0\}.$$

(3) For the first case, we have

$$\begin{aligned} \chi_{(\liminf A_i)}(x) = 1 &\iff x \in \liminf_i A_i \iff x \in A_i \text{ for all but finitely many } i \\ &\iff \chi_{A_i}(x) = 1 \text{ for all but finitely many } i \\ &\iff \text{there exists } i^* \text{ s.t. } \chi_{A_i}(x) = 1 \text{ for every } i \geq i^* \\ &\iff \liminf_i \chi_{A_i}(x) = 1. \end{aligned}$$

Since the two functions take values only in $\{0, 1\}$, the equivalence above implies that they are equal for every x . Therefore,

$$\chi_{(\liminf A_i)}(x) = \liminf_i \chi_{A_i}(x).$$

For the second case, we have

$$\begin{aligned}
\chi_{(\limsup A_i)}(x) = 1 &\iff x \in \limsup_i A_i \iff x \in A_i \text{ for infinitely many } i \\
&\iff \chi_{A_i}(x) = 1 \text{ for infinitely many } i \\
&\iff \sup_{i \geq i^*} \chi_{A_i}(x) = 1 \text{ for every } i^* \\
&\iff \limsup_i \chi_{A_i}(x) = 1.
\end{aligned}$$

Therefore,

$$\chi_{(\limsup A_i)}(x) = \limsup_i \chi_{A_i}(x).$$

Part II

Chapter 3

14 Exercise 3.1

14.1 Solution

Let $\{B_i\}$ be a pairwise disjoint sequence of sets in $\mathcal{A}$ and define

$$A_i = \bigcup_{k=1}^i B_k.$$

Then A_i is an increasing sequence of sets in $\mathcal{A}$ and

$$\bigcup_i A_i = \bigcup_i B_i.$$

Thus,

$$\mu\left(\bigcup_i B_i\right) = \mu\left(\bigcup_i A_i\right) = \lim_{i \rightarrow \infty} \mu(A_i) = \lim_{i \rightarrow \infty} \mu\left(\bigcup_{k=1}^i B_k\right).$$

Since μ is finitely additive, we have

$$\lim_{i \rightarrow \infty} \mu\left(\bigcup_{k=1}^i B_k\right) = \lim_{i \rightarrow \infty} \sum_{k=1}^i \mu(B_k) = \sum_{i=1}^{\infty} \mu(B_i).$$

Hence μ is countably additive, and therefore a measure.

15 Exercise 3.2

15.1 Solution

Let $\{B_i\}$ be a pairwise disjoint sequence of sets in $\mathcal{A}$ and define

$$A_i = \bigcup_{k=i+1}^{\infty} B_k.$$

Then $\{A_i\}$ is a decreasing sequence of sets in $\mathcal{A}$ and

$$A_i \downarrow \emptyset.$$

Thus, by assumption,

$$\mu(A_i) \rightarrow 0.$$

Since μ is finitely additive, for every $n \in \mathbb{N}$, we have

$$\mu\left(\bigcup B_i\right) = \mu\left(\left(\bigcup_{i=1}^n B_i\right) \cup A_n\right) = \mu(A_n) + \sum_{i=1}^n \mu(B_i).$$

Therefore,

$$\sum_{i=1}^{\infty} \mu(B_i) = \lim_{n \rightarrow \infty} \sum_{i=1}^n \mu(B_i) = \mu\left(\bigcup B_i\right) - \lim_{n \rightarrow \infty} \mu(A_n) = \mu\left(\bigcup B_i\right).$$

Hence μ is countably additive, and therefore a measure.

16 Exercise 3.3

16.1 Solution

Clearly, we have $\mu(\emptyset) = 0$.

Next, let $\{A_i\}$ be a pairwise disjoint sequence of sets in $\mathcal{A}$. We first show that at most one of the sets A_i is uncountable.

Suppose, to the contrary, that A_1 and A_2 are uncountable. Since $A_1 \cap A_2 = \emptyset$, we have

$$A_1 \subset A_2^c.$$

Because A_2 is uncountable and $A_2 \in \mathcal{A}$, its complement A_2^c is countable. Hence A_1 is countable, a contradiction.

If every A_i is countable, then $\bigcup_i A_i$ is countable. Therefore,

$$\mu\left(\bigcup_i A_i\right) = 0, \quad \sum_i \mu(A_i) = 0.$$

Otherwise, exactly one of the A_i is uncountable. Then $\bigcup_i A_i$ is uncountable, and

$$\mu\left(\bigcup_i A_i\right) = 1, \quad \sum_i \mu(A_i) = 1.$$

Hence μ is countably additive, and therefore a measure.

17 Exercise 3.4

17.1 Solution

$$\begin{aligned}\mu(A) + \mu(B) &= \mu((A \cap B) \cup (A \setminus B)) + \mu(B) \\ &= \mu(A \cap B) + \mu(A \setminus B) + \mu(B) \\ &= \mu(A \cap B) + \mu((A \setminus B) \cup B) \\ &= \mu(A \cap B) + \mu(A \cup B).\end{aligned}$$

18 Exercise 3.5

18.1 Solution

First, we have $\sum_n a_n \mu_n(\emptyset) = 0$, since

$$\mu_1(\emptyset) = \mu_2(\emptyset) = \cdots = 0.$$

Next, let $\{A_i\}$ be a pairwise disjoint sequence of measurable sets. Then

$$\sum_n a_n \mu_n \left(\bigcup_i A_i \right) = \sum_n a_n \left(\sum_i \mu_n(A_i) \right) = \sum_n \sum_i a_n \mu_n(A_i) = \sum_i \sum_n a_n \mu_n(A_i),$$

where the last equality follows from $a_n \mu_n(A_i) \geq 0$.

Hence $\sum_n a_n \mu_n$ is countably additive, and therefore a measure.

19 Exercise 3.6

19.1 Solution

First, we have $\nu(\emptyset) = \mu(\emptyset \cap B) = \mu(\emptyset) = 0$.

Next, let $\{A_i\}$ be a pairwise disjoint sequence of sets in $\mathcal{A}$. Then

$$\nu\left(\bigcup_i A_i\right) = \mu\left(\left(\bigcup_i A_i\right) \cap B\right) = \mu\left(\bigcup_i (A_i \cap B)\right) = \sum_i \mu(A_i \cap B) = \sum_i \nu(A_i).$$

Hence ν is countably additive, and therefore a measure.

20 Exercise 3.7

20.1 Solution

We will show that μ is a measure.

First, we have $\mu(\emptyset) = 0$, since

$$\mu(\emptyset) = \lim_{n \rightarrow \infty} \mu_n(\emptyset) = 0.$$

Next, let $\{A_i\}$ be a pairwise disjoint sequence of sets in $\mathcal{A}$. Then

$$\mu\left(\bigcup_i A_i\right) = \lim_{n \rightarrow \infty} \mu_n\left(\bigcup_i A_i\right) = \lim_{n \rightarrow \infty} \sum_{i=1}^{\infty} \mu_n(A_i).$$

1. Suppose that $\mu_n(A) \uparrow$ for each $A \in \mathcal{A}$. Let

$$c_{ni} = \mu_n(A_i).$$

Since c_{ni} is a non-negative real number that increases in n to $c_i = \mu(A_i)$ for each i , we have

$$\lim_{n \rightarrow \infty} \sum_{i=1}^{\infty} \mu_n(A_i) = \sum_{i=1}^{\infty} \mu(A_i).$$

Hence μ is countably additive, and therefore a measure.

2. Suppose that $\mu_n(A) \downarrow$ for each $A \in \mathcal{A}$ and $\mu_1(X) < \infty$. Let

$$c_{ni} = \mu_n(A_i).$$

Note that c_{ni} is a non-negative real number that decreases in n to $c_i = \mu(A_i)$ for each i . Thus, we have

$$\sup_n |c_{ni}| = \mu_1(A_i).$$

Since μ_1 is a measure, then

$$\sum_{i=1}^{\infty} \sup_n |c_{ni}| = \sum_{i=1}^{\infty} \mu_1(A_i) = \mu_1\left(\bigcup_i A_i\right) \leq \mu_1(X) < \infty.$$

Therefore, we have

$$\lim_{n \rightarrow \infty} \sum_{i=1}^{\infty} \mu_n(A_i) = \sum_{i=1}^{\infty} \mu(A_i).$$

Hence μ is countably additive, and therefore a measure.

21 Exercise 3.8

21.1 Solution

Let $\mathcal{C} = \{A \cup N : A \in \mathcal{A}, N \in \mathcal{N}\}$. First, we will show that $\mathcal{C}$ is a σ -algebra.

(1) $\emptyset, X \in \mathcal{C}$

Since $\emptyset \in \mathcal{A} \cap \mathcal{N}$, we have

$$\emptyset = \emptyset \cup \emptyset \in \mathcal{C}.$$

Also,

$$X = X \cup \emptyset \in \mathcal{C}.$$

(2) If $E \in \mathcal{C}$, then $E^c \in \mathcal{C}$

Let $E = A \cup N$, where $A \in \mathcal{A}$ and $N \in \mathcal{N}$. Then there exists $F \in \mathcal{A}$ such that

$$N \subset F, \quad \mu(F) = 0.$$

Thus, we have

$$E^c = A^c \cap N^c = A^c \cap (F^c \cup (F \setminus N)) = (A^c \cap F^c) \cup (A^c \cap (F \setminus N)).$$

Since $A, F \in \mathcal{A}$, we have $A^c \cap F^c \in \mathcal{A}$. Moreover, since

$$A^c \cap (F \setminus N) = (F \setminus N) \setminus A \subset F,$$

we have $A^c \cap (F \setminus N) \in \mathcal{N}$. Hence, $E^c \in \mathcal{C}$.

(3) If $E_1, E_2, \dots \in \mathcal{C}$, then $\bigcup_i E_i \in \mathcal{C}$

Let $E_i = A_i \cup N_i$, where $A_i \in \mathcal{A}$ and $N_i \in \mathcal{N}$. Then there exists $F_i \in \mathcal{A}$ such that

$$N_i \subset F_i, \quad \mu(F_i) = 0.$$

Since

$$\bigcup_i N_i \subset \bigcup_i F_i, \quad \mu\left(\bigcup_i F_i\right) \leq \sum_i \mu(F_i) = 0,$$

we have

$$\bigcup_i N_i \in \mathcal{N}.$$

Hence, we have

$$\bigcup_i E_i = \left(\bigcup_i A_i \right) \cup \left(\bigcup_i N_i \right) \in \mathcal{C}.$$

Therefore, $\mathcal{C}$ is a σ -algebra.

Next, we will show that $\mathcal{B} = \mathcal{C}$.

Let $E = A \cup N \in \mathcal{C}$. Then $A \in \mathcal{A}$ and $N \in \mathcal{N}$, which implies that $A, N \in \mathcal{B}$. Thus, we have $A \cup N \in \mathcal{B}$. Therefore,

$$\mathcal{C} \subset \mathcal{B}.$$

On the other hand, let $E \in \mathcal{A} \cup \mathcal{N}$. If $E \in \mathcal{A}$, then

$$E = E \cup \emptyset \in \mathcal{C}.$$

And if $E \in \mathcal{N}$, then

$$E = \emptyset \cup E \in \mathcal{C}.$$

Thus, we have

$$\mathcal{A} \cup \mathcal{N} \subset \mathcal{C}.$$

Since $\mathcal{C}$ is a σ -algebra containing $\mathcal{A} \cup \mathcal{N}$, we have

$$\mathcal{B} = \sigma(\mathcal{A} \cup \mathcal{N}) \subset \mathcal{C}.$$

Hence, $\mathcal{B} = \mathcal{C}$.

Finally, we will show that $\bar{\mu}$ is uniquely defined.

Suppose that

$$B = A_1 \cup N_1 = A_2 \cup N_2,$$

where $A_1, A_2 \in \mathcal{A}$ and $N_1, N_2 \in \mathcal{N}$. If $x \in A_1 \setminus A_2$, then

$$x \in A_1 \subset B = A_2 \cup N_2, \quad x \notin A_2 \quad \Rightarrow \quad x \in N_2.$$

Thus, we have

$$A_1 \setminus A_2 \subset N_2.$$

Similarly, we have

$$A_2 \setminus A_1 \subset N_1.$$

Since $N_2 \in \mathcal{N}$, there is a set $B_2 \in \mathcal{A}$ such that $N_2 \subset B_2$ and $\mu(B_2) = 0$. Moreover, since

$$A_1 \setminus A_2 \subset N_2 \subset B_2,$$

we have

$$\mu(A_1 \setminus A_2) = 0.$$

Similarly,

$$\mu(A_2 \setminus A_1) = 0.$$

Since $A_1 = (A_1 \cap A_2) \cup (A_1 \setminus A_2)$ and this union is disjoint, we have

$$\mu(A_1) = \mu(A_1 \cap A_2) + \mu(A_1 \setminus A_2) = \mu(A_1 \cap A_2).$$

Similarly, we have $\mu(A_2) = \mu(A_1 \cap A_2)$. Hence,

$$\mu(A_1) = \mu(A_2).$$

Hence,

$$\bar{\mu}(B) = \mu(A_1) = \mu(A_2),$$

so $\bar{\mu}$ is well-defined.

22 Exercise 3.9

22.1 Solution

First, we consider the case where m and n are finite.

Let

$$\mathcal{A} = \{A \in \mathcal{B} : m(A) = n(A)\}.$$

We will show that $\mathcal{A}$ is a monotone class.

1. If $A_i \uparrow A$ and each $A_i \in \mathcal{A}$, then

$$\begin{aligned} m(A) &= m\left(\bigcup_i A_i\right) = m\left(A_1 \cup \bigcup_{i=1}^{\infty} (A_{i+1} - A_i)\right) = m(A_1) + \sum_{i=1}^{\infty} m(A_{i+1} - A_i) \\ &= m(A_1) + \sum_{i=1}^{\infty} (m(A_{i+1}) - m(A_i)) = n(A_1) + \sum_{i=1}^{\infty} (n(A_{i+1}) - n(A_i)) \\ &= n\left(A_1 \cup \bigcup_{i=1}^{\infty} (A_{i+1} - A_i)\right) = n\left(\bigcup_i A_i\right) \\ &= n(A). \end{aligned}$$

Hence, $A \in \mathcal{A}$.

2. Suppose that $A_i \downarrow A$ and each $A_i \in \mathcal{A}$. Since

$$(-i, i) \uparrow \mathbb{R},$$

we have $\mathbb{R} \in \mathcal{A}$. Thus, if $E \in \mathcal{A}$, then

$$m(E^c) = m(\mathbb{R}) - m(E) = n(\mathbb{R}) - n(E) = n(E^c),$$

which implies that $E^c \in \mathcal{A}$. Thus, we have $A_i^c \in \mathcal{A}$.

Since $A_i^c \uparrow A^c$, Step 1 implies that $A^c \in \mathcal{A}$. Thus, we have $A \in \mathcal{A}$.

Therefore, $\mathcal{A}$ is a monotone class.

Next, we will show that $\mathcal{B} = \mathcal{A}$.

Let $\mathcal{C}$ be the collection of all finite disjoint unions of intervals of the forms

$$(a, b], \quad (-\infty, b], \quad (a, \infty),$$

where $a, b \in \mathbb{R}$, together with $\emptyset$ and $\mathbb{R}$. Then $\mathcal{C}$ is an algebra and

$$\sigma(\mathcal{C}) = \mathcal{B}.$$

First, for any $a < b$, we have

$$(a, b + 1/n) \downarrow (a, b].$$

Since each $(a, b + 1/n) \in \mathcal{A}$ by assumption and $\mathcal{A}$ is a monotone class, it follows that

$$(a, b] \in \mathcal{A}.$$

Next, for any $b \in \mathbb{R}$,

$$(-n, b] \uparrow (-\infty, b].$$

Since $(-n, b] \in \mathcal{A}$ for all sufficiently large n , the monotone class property gives

$$(-\infty, b] \in \mathcal{A}.$$

We know that $\mathcal{A}$ is closed under complements. Hence

$$(a, \infty) = (-\infty, a]^c \in \mathcal{A}.$$

Therefore, m and n agree on every interval used in the definition of $\mathcal{C}$. By finite additivity, they agree on every finite disjoint union of such intervals. Hence,

$$\mathcal{C} \subset \mathcal{A}.$$

Since $\mathcal{A}$ is a monotone class containing $\mathcal{C}$,

$$\mathcal{M}(\mathcal{C}) \subset \mathcal{A}.$$

By the monotone class theorem,

$$\mathcal{B} = \sigma(\mathcal{C}) = \mathcal{M}(\mathcal{C}),$$

and therefore

$$\mathcal{B} \subset \mathcal{A}.$$

Since $\mathcal{A} \subset \mathcal{B}$ by definition, we conclude that

$$\mathcal{A} = \mathcal{B}.$$

Thus,

$$m(A) = n(A) \quad \text{for every } A \in \mathcal{B}.$$

Now suppose that at least one of m and n is not finite. For $k \in \mathbb{N}$, define

$$m_k(A) = m(A \cap (-k, k)), \quad n_k(A) = n(A \cap (-k, k)).$$

Thus m_k and n_k are finite, since

$$m((-k, k)) = n((-k, k)) < \infty.$$

For any open interval (a, b) , we know that $(a, b) \cap (-k, k)$ is an open interval or the emptyset. Thus,

$$m_k((a, b)) = m((a, b) \cap (-k, k)) = n((a, b) \cap (-k, k)) = n_k((a, b)).$$

Hence, from finite case, we can say that $m_k(A) = n_k(A)$ for any $A \in \mathcal{B}$.

For any $A \in \mathcal{B}$, defined

$$A_k = A \cap (-k, k).$$

Then we have

$$A_k \uparrow A.$$

Since

$$m(A_k) = n(A_k)$$

for every k , and

$$A = A_1 \cup \bigcup_{k=2}^{\infty} (A_k \setminus A_{k-1}),$$

we obtain

$$m(A) = m(A_1) + \sum_{k=2}^{\infty} m(A_k \setminus A_{k-1}) = n(A_1) + \sum_{k=2}^{\infty} n(A_k \setminus A_{k-1}) = n(A).$$

23 Exercise 3.10

23.1 Solution

The statement is false, even when m and n are finite measures.

Consider

- (1) $X = \{1, 2, 3\}$
- (2) $\mathcal{A} = \mathcal{P}(X)$
- (3) $\mathcal{C} = \{\{1, 2\}, \{2, 3\}\}$.

Then $\sigma(\mathcal{C}) = \mathcal{A}$.

Define m , where

- (1) $m(\{\emptyset\}) = 0, m(\{1\}) = m(\{3\}) = 1, m(\{2\}) = 0,$
- (2) $m(\{1, 2\}) = m(\{2, 3\}) = 1, m(\{1, 3\}) = 2,$
- (3) $m(\{1, 2, 3\}) = 2.$

Define n , where

- (1) $n(\{\emptyset\}) = 0, n(\{1\}) = n(\{3\}) = 0, n(\{2\}) = 1,$
- (2) $n(\{1, 2\}) = m(\{2, 3\}) = 1, n(\{1, 3\}) = 0,$
- (3) $n(\{1, 2, 3\}) = 1.$

Then m, n are finite measures (also σ -finite measures), agree on $\mathcal{C}$, but do not agree on $\sigma(\mathcal{C})$.

Part III

Chapter 4

24 Exercise 4.1

24.1 Problem

Paraphrase the problem here.

24.2 Solution

Suppose that

$$A_n \uparrow A.$$

Then

$$\mu(A) = \lim_{n \rightarrow \infty} \mu(A_n).$$

Hence the result follows.

25 Exercise 4.2

25.1 Problem

Paraphrase the problem here.

25.2 Solution

Suppose that

$$A_n \uparrow A.$$

Then

$$\mu(A) = \lim_{n \rightarrow \infty} \mu(A_n).$$

Hence the result follows.

26 Exercise 4.3

26.1 Problem

Paraphrase the problem here.

26.2 Solution

Suppose that

$$A_n \uparrow A.$$

Then

$$\mu(A) = \lim_{n \rightarrow \infty} \mu(A_n).$$

Hence the result follows.

References